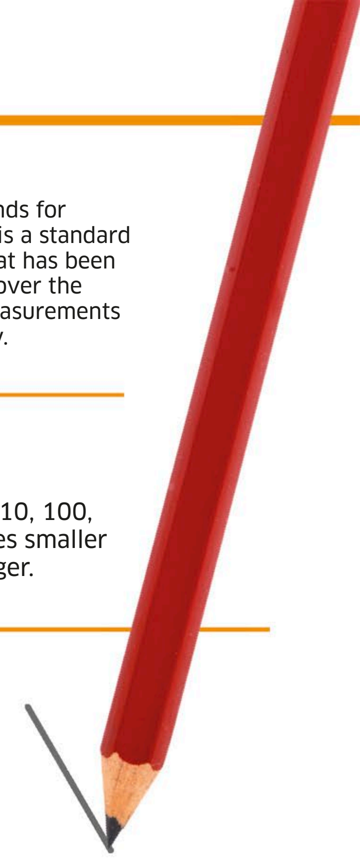
Units of measurement

Scientists measure quantities—such as length, mass, or time—using numbers, so that their sizes can be compared. For each kind of quantity, these measurements must be in units that mean the same thing wherever in the world the measurements are made.

SI units
The abbreviation “SI” stands for *Système International*. It is a standard system of metric units that has been adopted by scientists all over the world so that all their measurements are done in the same way.



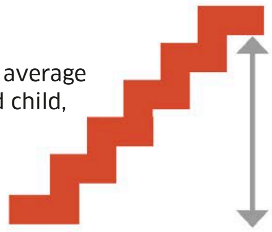
Base quantities

Just seven quantities give the most basic information about everything around us. Each is measured in SI units and uses a symbol as an abbreviation. The SI system is metric, meaning that smaller and larger units are obtained by dividing or multiplying by 10, 100, 1,000, etc. Centimeters, for instance, are 100 times smaller than a meter, but kilometers are 1,000 times bigger.

LENGTH

SI unit: meter (m)

One meter is about the average height of a 3½-year-old child, or five steps up a typical staircase.



- A millionth of a meter (1 micrometer) = the length of a bacterium.
- A thousandth of a meter (1 millimeter) = the diameter of a pinhead.
- 1,000 meters (1 kilometer) = the average distance an adult walks in 12 minutes.

MASS

SI unit: kilogram (kg)

One kilogram is the mass of one liter of water, or about the mass of an average-sized pineapple.

- A thousand trillionth of a kilogram (1 picogram) = the mass of a bacterium.
- A thousandth of a kilogram (1 gram) = the mass of a paper clip.
- 1,000 kilograms (1 metric ton) = the average mass of an adult walrus.

TIME

SI unit: second (s)

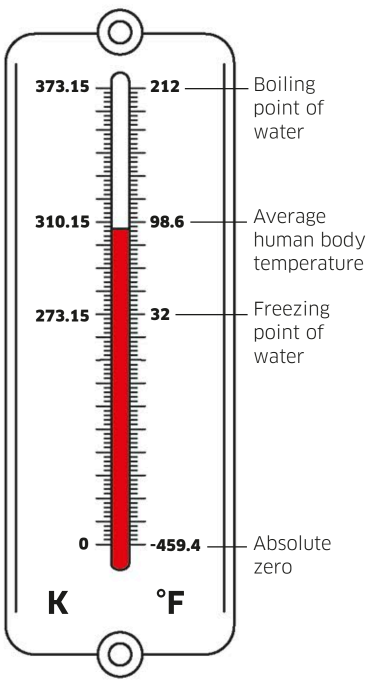
One second is the time it takes to swallow a mouthful of food, or to write a single-digit number.

- A thousandth of a second (1 millisecond) = the time taken by the brain to fire a nerve impulse.
- A tenth of a second (1 decisecond) = a blink of an eye.
- 1 billion seconds (1 gigasecond) = 32 years.

TEMPERATURE

SI unit: kelvin (K)

Just one degree rise in temperature can make you feel hot and feverish.



Temperature scales
In the USA, an everyday temperature scale uses degrees Fahrenheit (°F), where the freezing point of water is 32°F. Kelvin measures all the way down to absolute zero—where heat energy does not exist.

- 0 kelvin = absolute zero, when all objects and their particles are still.
- 1 kelvin = the coldest known object in the universe, the Boomerang Nebula.
- 1,000 kelvin = the temperature inside a charcoal fire.

ELECTRICAL CURRENT

SI unit: ampere (A)

One ampere is about the current running through a 100 W light bulb.

- A thousandth of an ampere (1 milliampere) = the current in a portable hearing aid.
- 100,000 amperes = the current in the biggest lightning strikes.
- 10 thousand billion amperes = the current in the spiral arms of the Milky Way.

LIGHT INTENSITY

SI unit: candela (cd)

One candela is the light intensity given off by a candle flame.

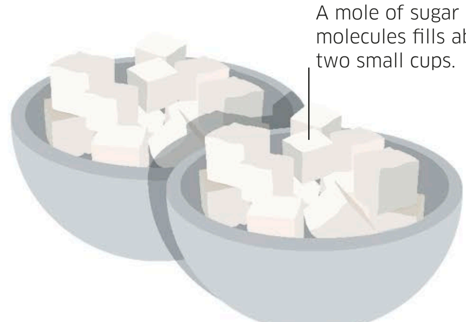
- A millionth of a candela = the lowest light intensity perceived by human vision.
- A thousandth of a candela = a typical night sky away from city lights.
- 1 billion candelas = the intensity of the sun when viewed from Earth.

AMOUNT OF A SUBSTANCE

SI unit: mole (mol)

One mole is a set number of atoms, molecules, or other particles. Because substances all have different atomic structures, one mole of one substance may be very different to that of another.

A mole of gold atoms is in about six gold coins.



A mole of sugar molecules fills about two small cups.

- A tenth of a mole of iron atoms = the amount of iron in the human body.
- 1,000 moles of carbon atoms = the amount of carbon in the human body.
- 10 million trillion moles of oxygen molecules = the amount of oxygen in Earth’s atmosphere.